Yashwardhan Panchal
CSCI 5561 - Computer Vision
October 01, 2026

HW1 Report

Assignment item	**Points**	**Location in report**	**Extra credit?**
Report Quality	5	-	-
Image Capture and Feature Extraction	10	Section 1, p. 2	-
Descriptor Matching	10	Section 2, p. 3	-
Homography Estimation	10	Section 3, p. 4	-
Basic Robust Estimation with RANSAC	20	Section 4, p. 5	-
Warping and Compositing	15	Section 5, p. 6	-
Feather Blending	10	Section 6, p. 8	-
Confidence-Based RANSAC Stopping	10	Section 7, p. 9	-
Exposure and Gain Compensation	10	Section 8, p. 9	-
Cylindrical Panorama	10	Appendix A, p. 11	Yes
Least Squares versus RANSAC	5	Appendix B, p. 12	Yes
Interpolation Comparison	5	Appendix C, p. 13	Yes
Total attempted before extra-credit conversion	**120**		
Maximum possible score after conversion	**110**		

Introduction

This report describes a two-image stitching pipeline: SIFT keypoints, nearest-neighbor descriptor matching with Lowe's ratio test, homography estimation with the normalized DLT, RANSAC, and inverse warping with compositing. SIFT detection and pixel sampling use OpenCV; matching, homography estimation, and RANSAC are my own implementations. In each image pair, Image 1 is the reference and the homography H maps Image 1 (source) to Image 2 (destination).

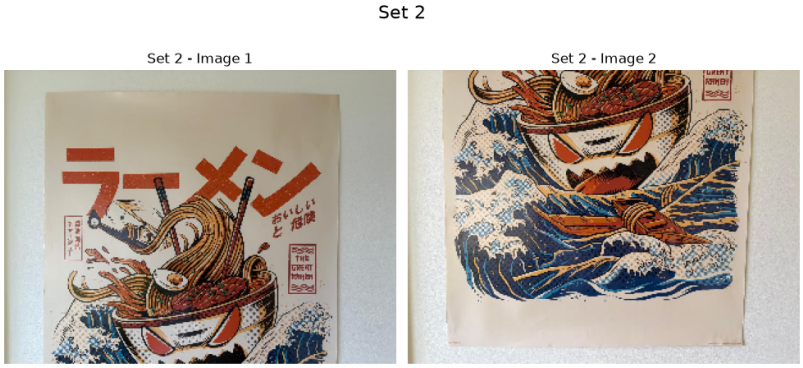
1 Image Capture and Feature Extraction

Set 1 (lab scene). The camera stayed at a fixed position and only changed orientation between the two images. The scene contains objects at different depths. A homography is valid because, for a pure camera rotation, the mapping between the two images is a homography regardless of scene depth.

Set 2 (poster). The camera was translated vertically with a fixed orientation. The scene is a poster on a wall, which is a single plane, and the mapping between two views of a plane is a homography even when the camera translates.

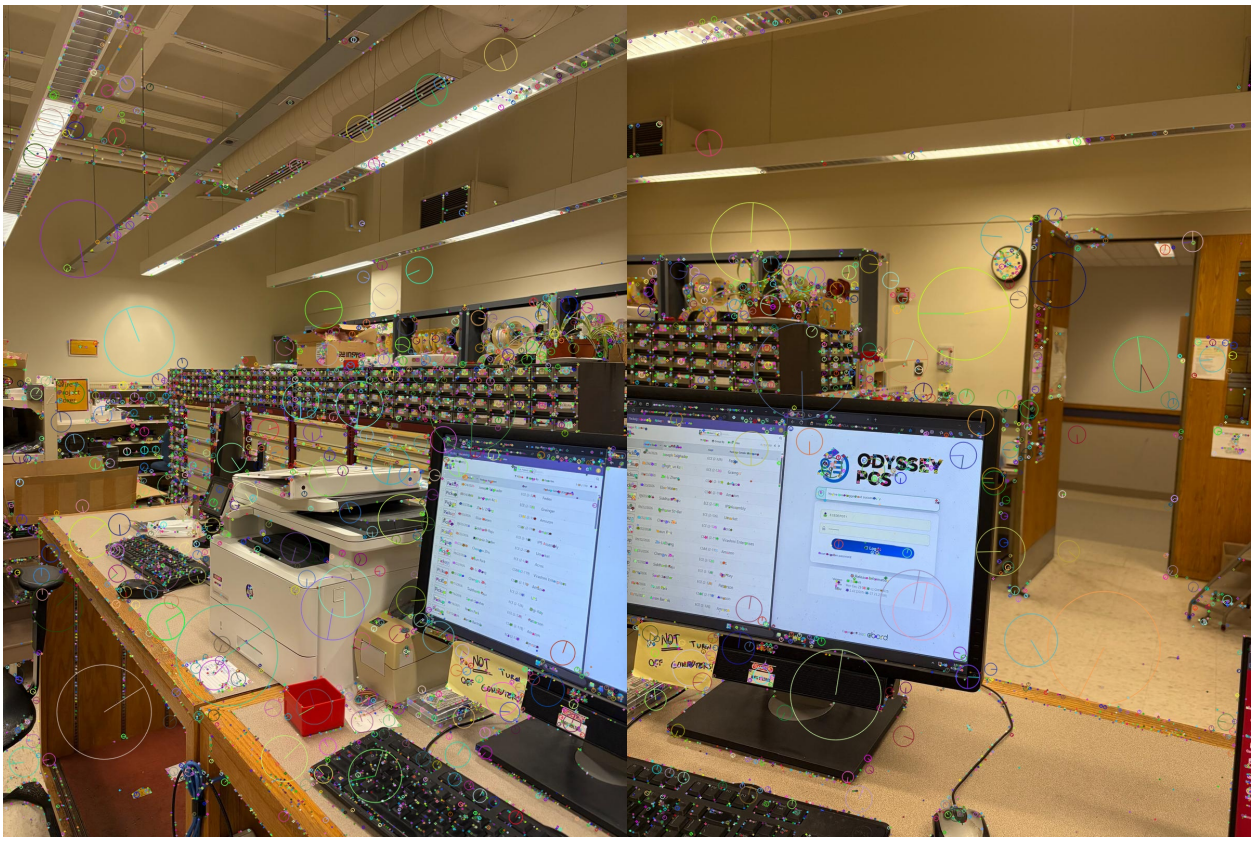
All images were taken with the primary camera of an iPhone 17. I extracted SIFT keypoints and descriptors with OpenCV using the default parameters. SIFT found 7460 and 3769 keypoints in the two Set 1 images and 9081 and 17587 in the two Set 2 images (Figure 1.2; circles show scale and orientation).

(a) Set 1 input images

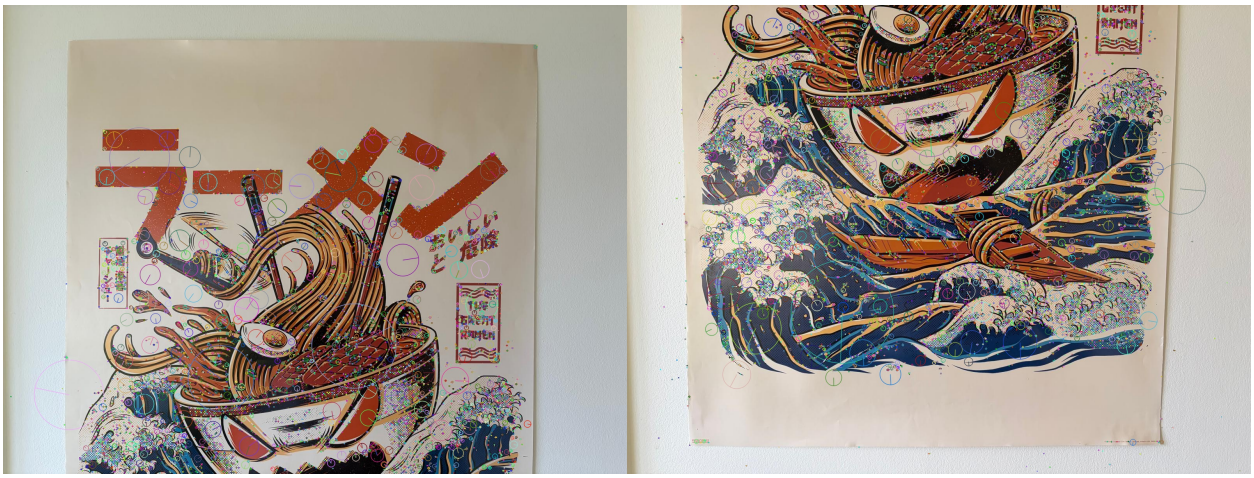


(b) Set 2 input images

Figure 1.1 Input images for both sets



(a) Set 1 keypoints



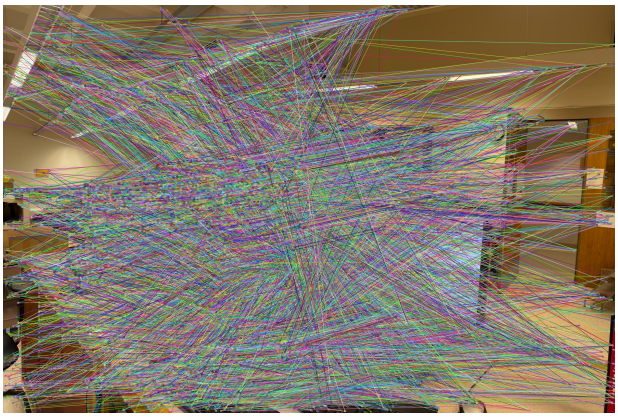
(b) Set 2 keypoints

Figure 1.2 SIFT keypoints

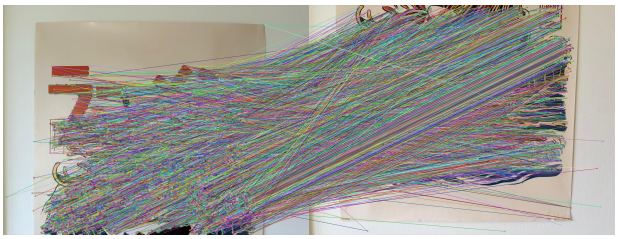
2 Descriptor Matching

For every descriptor in Image 1, I compute its Euclidean distance to all descriptors in Image 2 and keep the two nearest neighbors. The nearest neighbor is the candidate match, so there is one candidate per Image 1 keypoint (7460 for Set 1, 9081 for Set 2). Lowe's ratio test keeps a candidate only if its nearest distance is below 0.7 times the second-nearest distance, which removes

ambiguous matches. A threshold of 0.7 is stricter than the 0.8 suggested by Lowe, so it keeps fewer but more reliable matches. The test left 518 matches for Set 1 and 2464 for Set 2 (Figure 2.1).



(a) Set 1 before ratio test



(b) Set 2 before ratio test

(c) Set 1 after ratio test

(d) Set 2 after ratio test

Figure 2.1 Matches before (top) and after (bottom) the ratio test for Set 1 (left) and Set 2 (right)

3 Homography Estimation

The normalized DLT translates each point set to zero mean and scales it to a mean distance of $\sqrt{2}$ from the origin. It then solves the resulting $2N \times 9$ system with the SVD (smallest right singular vector), denormalizes the result, and scales it so $H_{33} = 1$. The same function handles four or more points.

I tested it with correspondences generated by projecting source points through the known homography

$$H = \begin{bmatrix} 0.9 & 0.1 & 40 \\ -0.05 & 1.1 & 20 \\ 3 \times 10^{-4} & -2 \times 10^{-4} & 1 \end{bmatrix},$$

in a 640×480 frame. The first test uses four manually chosen points near the frame corners, with

no three collinear. The second uses 20 points sampled uniformly at random over the frame. I also added Gaussian noise ($\sigma = 0.5$ px) to the destination points of the 20-point set. Errors are the mean distance between mapped and target points, because homographies are defined only up to scale and raw matrix entries are not comparable.

Test	**Points**	**Mean reprojection error (px)**
Noise-free	4	1.30×10^{-13}
Noise-free	20	1.17×10^{-13}
Noisy, vs. noisy points	20	0.62
Noisy, vs. clean points	20	0.30

Table 1 Homography recovery on synthetic correspondences.

4 Basic Robust Estimation with RANSAC

RANSAC estimates the Image 1 to Image 2 homography from the ratio-test matches. Each iteration (1) samples four distinct matches, skipping samples with any three collinear points in either image; (2) fits a homography with the normalized DLT; and (3) projects all Image 1 points into Image 2 and marks matches within 5 px (Euclidean) as inliers. The hypothesis with the most inliers is kept and refit with the normalized DLT on all of its inliers. The random generator is seeded for reproducibility. The 5 px threshold absorbs SIFT localization and matching noise. I used 1,000 iterations; for the lowest inlier ratio observed (0.36), 99% confidence needs only about 287 samples (Section 7), so this leaves ample margin.

Set 1 has 187 inliers out of 518 matches (ratio 0.36) and Set 2 has 2,316 out of 2,464 (ratio 0.94); Figure 4.1 shows inliers in green and outliers in red. The mean inlier error fell from 1.48 to 0.87 px (Set 1) and from 1.72 to 1.07 px (Set 2) after the refit, with both homographies evaluated on the same inlier set. Before the refit, the homography comes from only the four sampled matches. It fits those four points exactly but is sensitive to their localization noise, while the refit averages that noise over all inliers. The DLT minimizes an algebraic rather than a geometric error, so a decrease is not guaranteed, but it occurred in both sets.

(a) Set 1

(b) Set 2

Figure 4.1 RANSAC matches (inliers green, outliers red)

5 Warping and Compositing

Image 1 is the reference. Because H maps Image 1 to Image 2, I invert it to bring Image 2 into Image 1's frame. The canvas bounds are the floor of the minimum and the ceiling of the maximum of Image 1's corners and Image 2's transformed corners, and a translation by the negative minimum keeps all coordinates non-negative. The canvases are 3139×3064 px (Set 1) and 1605×1981 px (Set 2).

Warping uses inverse mapping: each canvas pixel is mapped back into the source image by the inverse transform (followed by the homogeneous divide) and sampled with bilinear interpolation, using `cv2.remap` with `cv2.INTER_LINEAR` on the coordinates I computed. A canvas pixel is valid if its source location lies inside the image. Compositing is overwrite: Image 1 is written first, then Image 2 wherever it is valid, which leaves a visible seam.

To verify the warp, I used a synthetic 100×80 image with three white landmarks at `[20, 20]`, `[70, 25]`, and `[35, 60]`, warped onto a 180×140 canvas with a canvas offset of $(30, 25)$

(Figure 5.1).

- Translation, $\begin{bmatrix} 1 & 0 & -20 \\ 0 & 1 & -15 \\ 0 & 0 & 1 \end{bmatrix}$: with the offset, the effective map is $(x, y) \mapsto (x + 10, y + 10)$, so the landmarks move to `[30, 30]`, `[80, 35]`, `[45, 70]`.
- Scale, $\begin{bmatrix} 1.4 & 0 & 0 \\ 0 & 1.4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$: with the offset, the effective map is $(x, y) \mapsto (1.4x + 30,\ 1.4y + 25)$, so the landmarks move to `[58, 53]`, `[128, 60]`, `[79, 109]`.

The warped landmarks land at these predicted positions.

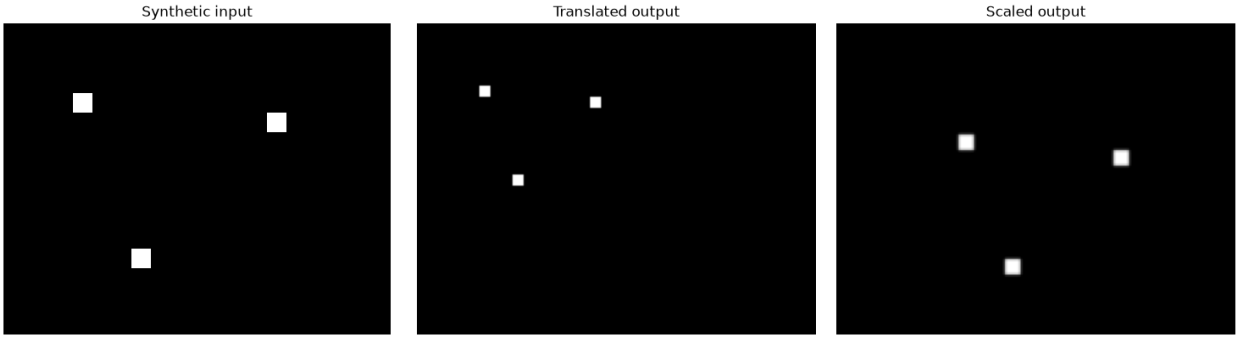
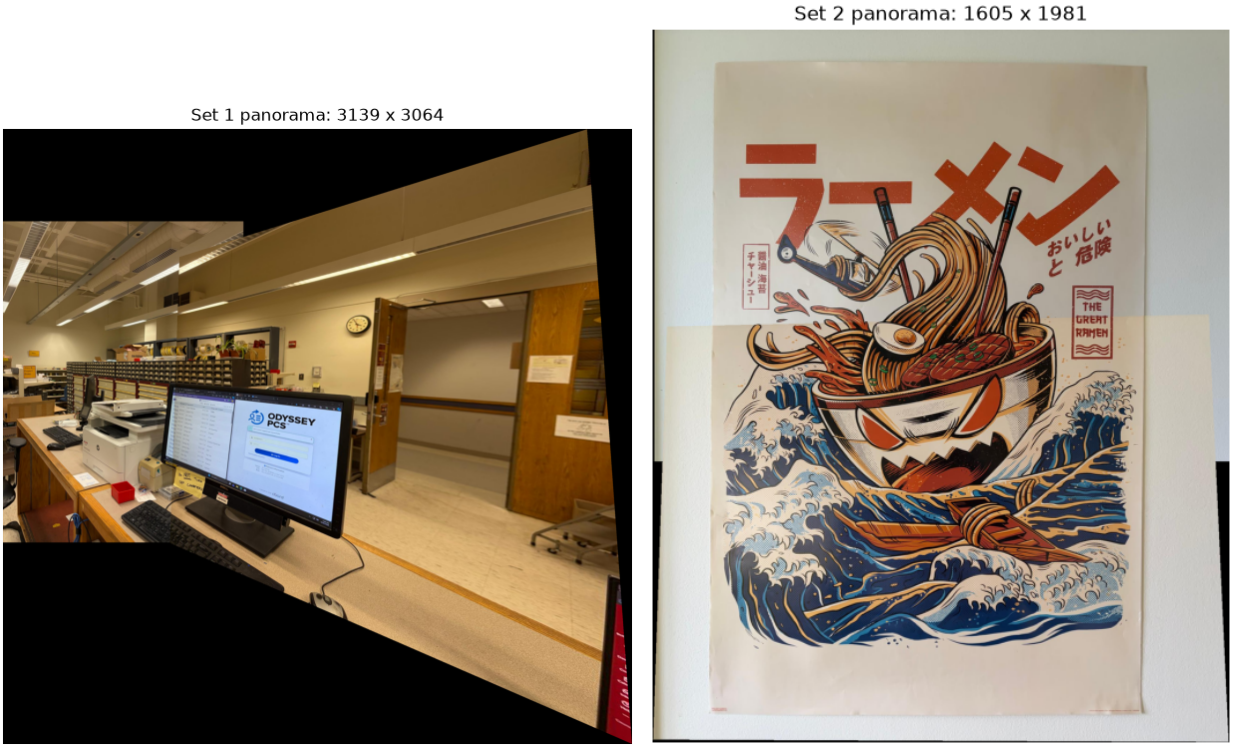


Figure 5.1 Synthetic warp tests: input, translated output, and scaled output



(a) Set 1 panorama *(b) Set 2 panorama*

Figure 5.2 Final panoramas (overwrite compositing, Image 1 as reference)

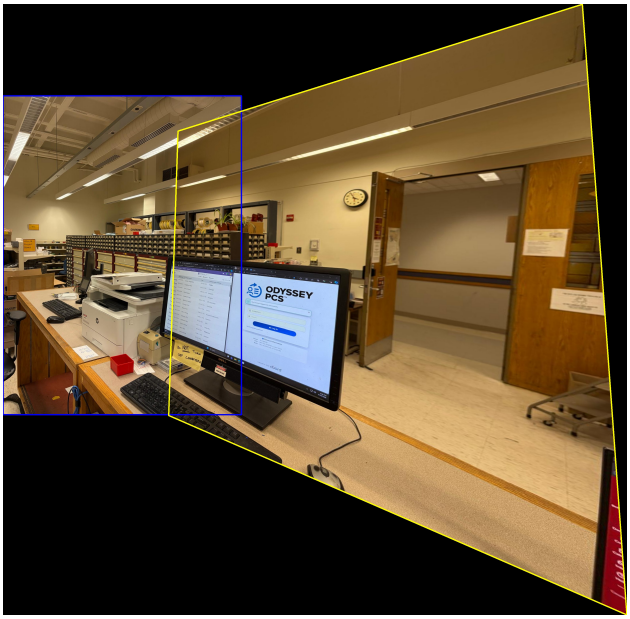


Figure 5.3 Set 1 annotated panorama: Image 1 is the reference (blue boundary) and Image 2 is warped into its frame (yellow boundary). Both boundaries are the input-image corners mapped with the same transforms used for compositing.

6 Feather Blending

For Set 2, overwrite compositing leaves a hard seam where Image 2 replaces Image 1. For feathering, each warped image's weight at a pixel is its distance to the nearest invalid pixel (zero outside the image, growing toward its interior), and each panorama pixel is the weighted average of the warped images. Contributions therefore change smoothly across the overlap. Figure 6.1 shows the weight map of Image 2 with the overwrite and feather results, and Figure 6.2 compares the same overlap crop. Feathering reduced abrupt intensity changes and made the seam less visible. Some ghosting or doubled edges remain because blending does not correct alignment errors.

Figure 6.1 Feather blend on Set 2: weight map of Image 2 (left), overwrite (middle), and feather (right)

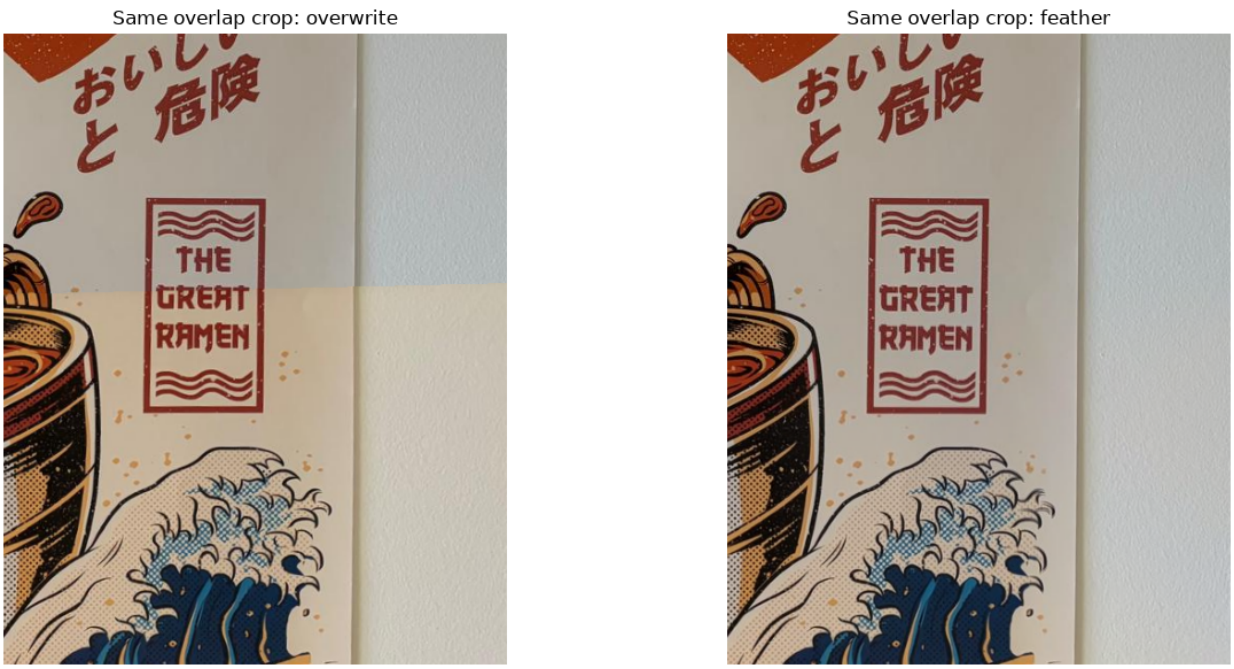


Figure 6.2 Same overlap crop under overwrite (left) and feather (right)

7 Confidence-Based RANSAC Stopping

Instead of a fixed 1,000 iterations, the required number of iterations is updated whenever a better hypothesis is found:

$$N = \left\lceil \frac{\log(1-p)}{\log(1-w^4)} \right\rceil,$$

where $p = 0.99$ is the target confidence and w is the best inlier fraction so far (w^4 is the probability that a four-point sample is all inliers). RANSAC stops when the iterations performed reach N, capped at 1,000.

For Set 1, adaptive RANSAC used 287 iterations and found 184 inliers, versus 187 with the fixed budget. For Set 2, it used 16 iterations and found 1,743 inliers, versus 2,316. The adaptive method is far cheaper, but stopping early can miss a better model that the longer run finds, as in Set 2, where it found 573 fewer inliers.

8 Exposure and Gain Compensation

I photographed the same scene at two different exposures (Figure 8.1) and aligned them with SIFT, ratio-test matching, and RANSAC as above, with Image 1 as the reference. Image 1's gain is fixed at $g_1 = 1$. Image 2's scalar gain is the least-squares estimate from corresponding pixels in the

warped overlap,

$$g_2 = \frac{\sum I_1 I_2}{\sum I_2^2},$$

where I_1 is the reference intensity and I_2 is the warped Image 2 intensity. The estimated $g_2 = 0.45$ shows that Image 2 was substantially brighter than Image 1.

Figure 8.2 compares the results. Without correction, the overlap shows a clear brightness discontinuity. After applying the gain, the two exposures are more consistent, and feather blending of the corrected images softens the remaining seam. Some small ghosting remains because feathering reduces intensity differences but does not correct geometric misalignment.

Figure 8.1 Exposure-mismatched input images

Figure 8.2 Panoramas before gain correction, after gain correction, and after gain correction with feather blending

Appendix A. Cylindrical Panorama

I captured three overlapping photographs in sequence while pivoting from approximately one position (Figure A.1). For the flat panorama, Image 2 is the reference. For the cylindrical panorama, each image is projected onto a cylinder with focal scale $f = 700$ px by inverse mapping: the source location of each cylinder pixel (x, y) is

$$x_c = f \tan\left(\frac{x - f_x}{f}\right) + f_x, \qquad y_c = \frac{y - f_y}{\cos\left((x - f_x)/f\right)} + f_y,$$

where (f_x, f_y) is the image center. The flat panorama is 3781×2458 px and the cylindrical one is 2026×1617 px (Figure A.2).

The flat result aligns the feature correspondences more accurately, so it shows less ghosting and cleaner seams. However, it stretches strongly under projection and has irregular boundaries across the wide field of view. The cylindrical result preserves the wide field of view more naturally and reduces stretching near the sides. Its remaining artifacts are ghosted building edges and people, vertical drift, exposure differences, and imperfect overlap transitions. These come mainly from parallax and from the camera not rotating exactly about its optical center. Feather blending softened the seams but cannot remove geometric misalignment.

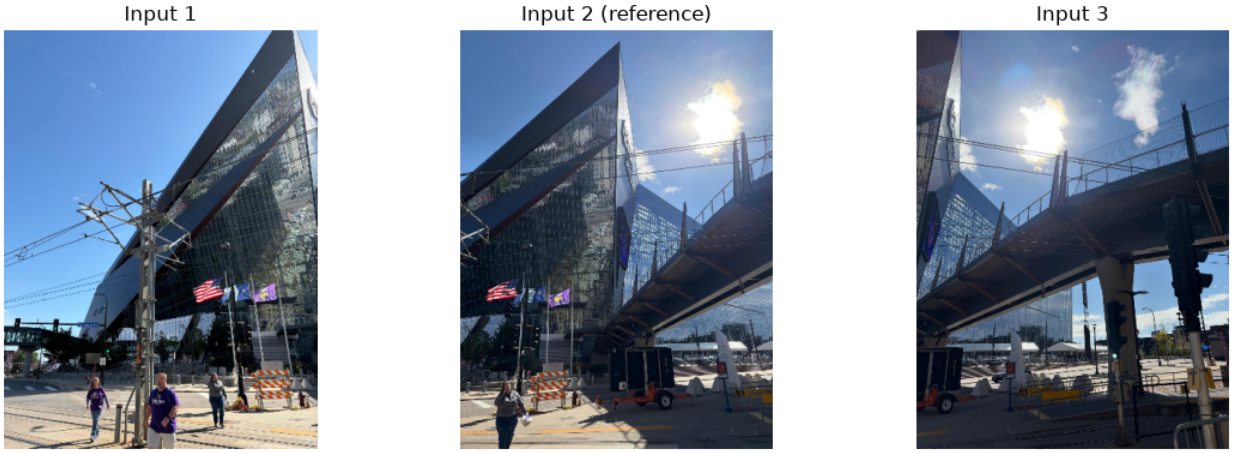


Figure A.1 Cylindrical input images

Figure A.2 Flat (left) and cylindrical (right) panoramas

Appendix B. Least Squares versus RANSAC

For Set 1, I compared a homography fit by least squares to all 518 ratio-test matches, outliers included, with the RANSAC-refit homography from the 187 inliers. Both were used to build panoramas from the same image pair (Figure B.1). The all-match homography has a mean reprojection error of 536.28 px over all 518 matches and produces a visibly distorted panorama. The RANSAC homography has a mean error of 0.87 px over its inliers and produces a well-aligned panorama. The two errors are measured over different match sets, but the gap shows how strongly incorrect correspondences pull the all-match fit: least squares weights every match equally, while RANSAC excludes the outliers.

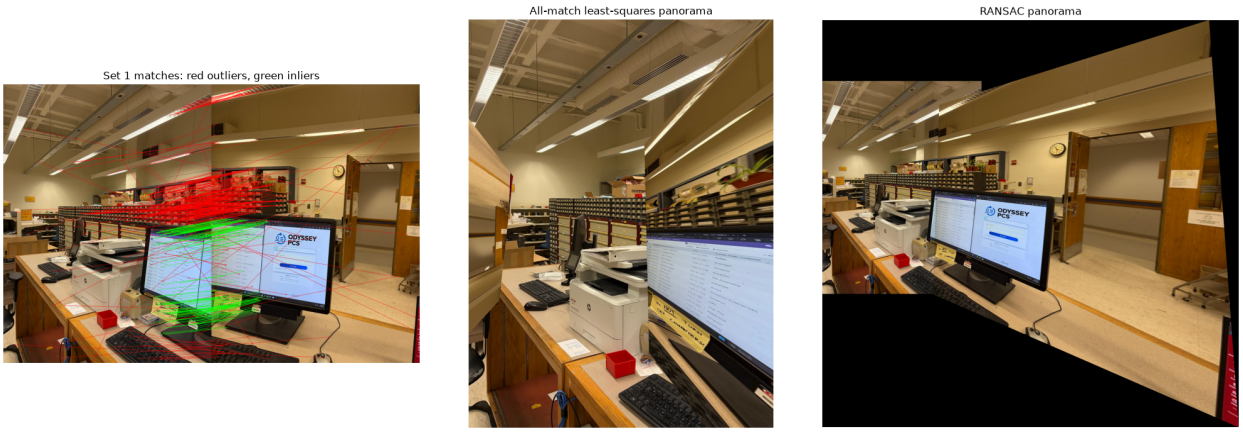


Figure B.1 Set 1 matches (inliers green, outliers red), all-match least-squares panorama, and RANSAC panorama

Appendix C. Interpolation Comparison

I rendered the Set 1 panorama with the same RANSAC homography using nearest-neighbor and bilinear sampling (Figure C.1). Figure C.2 shows the same 180×280 px overlap crop from each, enlarged $3\times$ for display, together with the absolute difference amplified $5\times$. Over this crop the mean absolute difference is 2.9 intensity levels and the maximum is 72, concentrated along the straight edges of the monitor screen. Nearest-neighbor sampling leaves hard pixel blocks and jagged edges, while bilinear interpolation averages neighboring samples and gives smoother edges with fewer block artifacts.

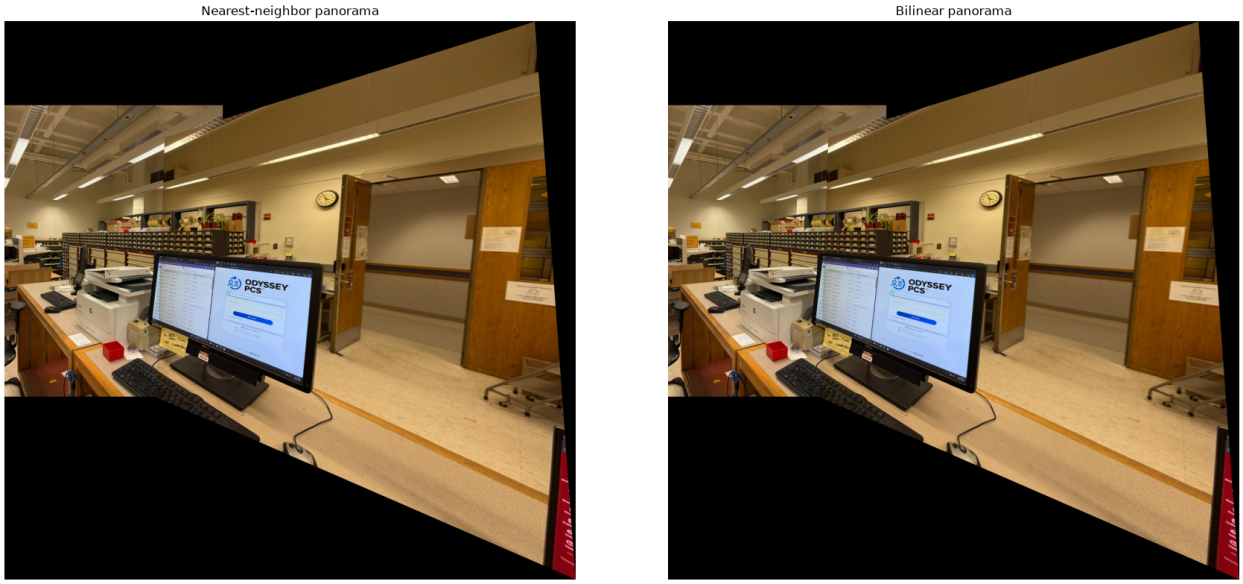


Figure C.1 Nearest-neighbor (left) and bilinear (right) panoramas

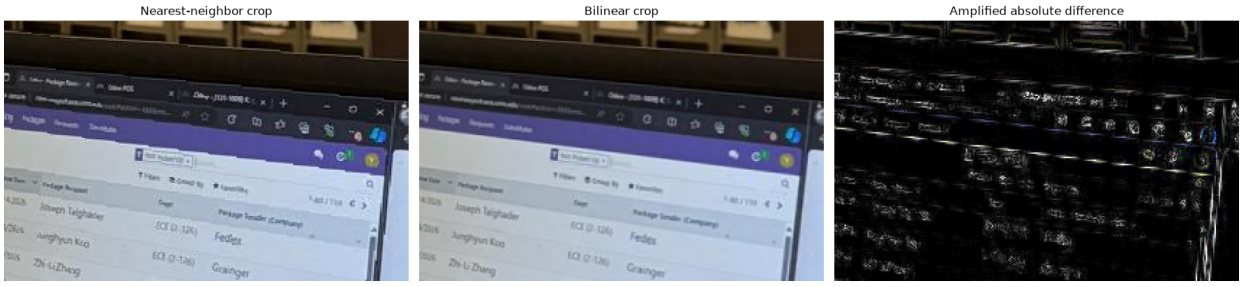


Figure C.2 Same crop under nearest-neighbor and bilinear sampling, and their amplified absolute difference